GoalConvo: Goal-Oriented Synthetic Dialogue Data Generation Using Multi-Agent LLM

Mrs. Tejaswini Zope
Computer Science and Engineering
Nutan Maharashtra Institute of Engineering and Technology
Pune, India.
tejaswinizope@gmail.com

Dr. Priyanka Patil
Nutan Maharashtra Institute of Engineering and Technology
Pune, India.
priyanka20.patil@gmail.com

Aditya Honkalas
Computer Science and Engineering
Nutan College of Engineering and Research
Pune, India.
adi.honkalas18@gmail.com

Emil Johny
Computer Science and Engineering
Nutan College of Engineering and Research
Pune, India.
emiljohny30@gmail.com

Aryan Maurya
Computer Science and Engineering
Nutan College of Engineering and Research
Pune, India.
108aryanmaurya@gmail.com

Prathamesh Podala
Computer Science and Engineering
Nutan College of Engineering and Research
Pune, India.
podalaprathamesh70@gmail.com

***Abstract* - The development of goal-oriented conversational AI systems relies on high-quality annotated datasets; however, creating these datasets is expensive, time-consuming, and challenging to scale across different domains. To help solve this problem, we introduce GoalConvo, a scalable framework for generating synthetic dialogues using multi-agent simulation and prompt-based orchestration. Our framework utilises an open-weight large language model to generate coherent, goal-driven conversations between simulated agents with predefined task objectives. GoalConvo features structured agent roles and modular goal definitions, enabling it to generate a wide range of interactions across various domains, including healthcare, technical support, and service booking. Users can adjust parameters for dialogue complexity, intent structure, and domain coverage. The approach is lightweight, reproducible, and does not need model fine-tuning or access to proprietary APIs. This method offers a practical and cost-effective way to generate synthetic datasets for training and evaluating goal-oriented dialogue systems.**

Keywords: Goal-oriented dialogue systems, Synthetic dialogue generation, Multi-agent simulation, Large language models, Task-oriented conversational AI, Prompt-based orchestration

I. INTRODUCTION

Task-oriented dialogue systems rely heavily on large-scale annotated conversational datasets; however, collecting such data through Wizard-of-Oz studies or crowdsourcing is both expensive and time-consuming. Benchmark datasets such as MultiWOZ have significantly contributed to the advancement of multi-domain task-oriented dialogue research, but they are affected by several limitations, including annotation noise, limited linguistic diversity, and insufficient coverage of rare or complex goals [1], [9], [13].

To mitigate these challenges, recent studies have investigated synthetic dialogue generation using large language models (LLMs). Multi-agent frameworks, such as ConvoGen, simulate interactions between role-conditioned agents to generate diverse conversational data [2], while AutoConv employs prompt-based techniques to produce information-seeking dialogues with minimal supervision [3]. Self-chat methods, including Baize, further demonstrate the ability of LLMs to generate coherent multi-turn dialogues by simulating both user and system roles [4]. Few current methods really aim at getting tasks done.

Most instead focus on broad topics or fact-based talk, drifting away from clear goals.

This study presents Goal Convo, a system designed to create purpose-driven conversations by anchoring each exchange to a specific user aim. Built using Mistral-7B - an openly available large language model - it orchestrates simulated talks between a User and a Support Bot via carefully crafted prompts. Rather than relying on training adjustments, it maintains alignment with objectives through structured guidance during generation. Every conversation reflects a clear intent, helping ensure tasks reach resolution as intended. Performance tests show effectiveness when compared with actual dialogue collections like MultiWOZ [1]..

II. RELATED WORK

Synthetic dialogue generation has gained significant attention as a practical solution to the high cost of collecting conversational data. Approaches such as ConvoGen build multi-agent systems in which LLM-based agents interact using predefined personas or situational prompts. AutoConv tackles data scarcity in question-answering settings by starting from a small seed dataset and expanding it through LLM-generated information-seeking dialogues. Similarly, self-chat methods (e.g., Xu et al., 2023) demonstrate that powerful chat models can generate both sides of a conversation from a single prompt. Together, these approaches have enabled the creation of large conversational datasets for training chatbots and instruction-following models. However, most existing methods focus on open-domain chat or question-answering scenarios, with limited attention to structured task completion.

GoalConvo takes a different approach by targeting task-oriented dialogues that are guided by explicit user goals. While it draws on the strengths of earlier synthetic data generation methods, it introduces goal conditioning and structured multi-agent orchestration to ensure that conversations remain focused on completing a specific task..

This work also builds on recent advances in large transformer-based language models, which have shown strong performance in few-shot text generation (e.g., GPT-3, T5). To evaluate the quality of the generated dialogues, we employ established semantic similarity metrics such as BERTScore [2], which help measure both fidelity and diversity. Overall, GoalConvo complements existing data augmentation efforts by enabling scalable generation of task-specific dialogue data for training and refining goal-oriented conversational AI systems.

III. LITREATURE SURVEY

Synthetic dialogue generation has become an increasingly attractive alternative to the costly and time-consuming process of collecting human-annotated conversational data. Early task-oriented dialogue systems commonly relied on Wizard-of-Oz studies, which led to the creation of benchmark datasets such as MultiWOZ and its subsequent versions that are still widely used for training and evaluation today [1], [9], [13]. Even though these collections of data helped push things forward, applying them elsewhere usually costs too much. They also tend to lack variety in how people speak.

New advances in big language systems now allow us to create chat examples automatically and in large amounts. Systems like ConvoGen build different kinds of talks by giving AI characters specific roles, then shaping their back-and-forth exchanges step by step [2]. In a related way, AutoConv targets conversations where someone looks for answers - using short prompts and slight adjustments to grow tiny starting sets into bigger ones [3]. A single model can play both speaker and listener using techniques like Baize or similar self-dialogue systems, needing almost no oversight [4], [11]. Even though they work well, such methods usually aim at general talk or answering questions instead of handling step-by-step tasks.

Looking at fake conversation data usually means using tools like BERTScore, since they track meaning more closely than old-style word overlap methods [5]. When it comes to systems built for specific tasks, how varied the words are matters just as much as whether the system finishes what it set out to do. Many existing

approaches do not explicitly enforce these properties, which can limit the usefulness of the generated data.

GoalConvo is designed to address these limitations by combining multi-agent prompting with explicit goal conditioning and automated quality checks. By grounding each dialogue in a clearly defined task, GoalConvo enables scalable generation of goal-oriented conversational data that is both diverse and directly applicable to training task-oriented dialogue systems.

IV. METHODOLOGY

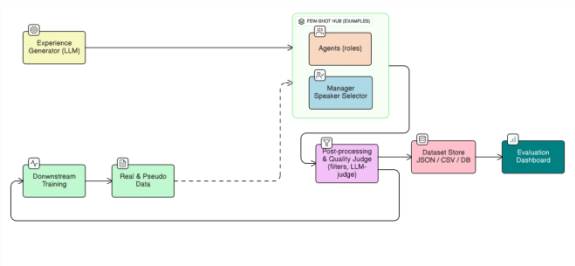

Fig. 4.1. Schematic of the GoalConvo data generation pipeline.

GoalConvo follows a structured pipeline in which a user experience generator first defines the goal and context of a dialogue. A multi-agent simulator, consisting of a User agent and a SupportBot agent, then carries out the conversation through iterative prompting. Finally, post-processing modules filter and evaluate the generated dialogue before storing the validated output in a dataset and presenting summary statistics on an evaluation dashboard

We implement GoalConvo as a multi-agent simulated environment built on Mistral-7B. As illustrated in Fig. 1, the framework consists of four main components: (A) Experience Generator, (B) Multi-Agent Simulator, (C) Post-Processing and Quality Judge, and (D) Dataset Store and Evaluation. Each component is described below.

1. **Experience Generator (Few-Shot Prompting):** Right away, the system picks a purpose for the conversation. Sometimes it pulls from set examples, other times from actual past messages, adding just enough background to start. With help from Mistral-7B and clever prompting, that thin outline grows into something more detailed. Imagine someone wanting to reserve a hotel room today - out comes a brief scenario plus the first thing that user might say. That starting point shapes everything after. The resulting output often includes meta-information such as the task objective, a brief user persona, and a conversation starter. To improve generation quality over time, we maintain a dynamically updated pool of few-shot examples. High-quality generated dialogues can be added back into this pool to guide future generations through iterative sampling, similar in spirit to ConvoGen's iterative hub. In practice, the generated goal and task description are prepended as system- or user-level instructions in the prompts provided to both agents.

2. **Multi-Agent Simulator (User and SupportBot Roles):** Using the output of the Experience Generator, the system simulates a dialogue by modeling the User and SupportBot as separate agents, both instantiated using Mistral-7B but conditioned with different role-specific prompts. Dialogue generation proceeds turn by turn. At each step, the User agent receives the full conversation history, including the task goal and the most recent SupportBot response, and generates the next user utterance aimed at progressing toward the goal. Now comes the part where SupportBot answers, shaped by what's just been said - clear, on point. Each prompt carries clear tags and guidance, so responses stay true to character, flow makes sense.

3. **Post-Processing & Quality Judge:** Once created, every conversation moves into cleanup to check standards. A series of basic rules scans for flaws like odd length, too much repeating, or broken logic - cutting out weak results. Then, another model steps in, judging whether replies fit the purpose, flow naturally, and finish what they started. If a chat falls short on any mark, it gets left behind. The ones that pass form the core of what ends up in the collection

4. **Dataset Store & Evaluation:** Stored conversations go into a labeled collection, complete with topic tags and purpose markers, making them easier to scale and access later. One tool checks how close meanings are, how varied the word choices appear, plus whether

goals were reached in each exchange. A display screen shows these measurements clearly, allowing consistent review and side by side looks at different dialogue sets.

$$P\big((u_t|H_{t-1}, g)\big) \; and \; P(s_t|H_t, g) \qquad [4.1]$$

Where:

H_t = dialogue history up to turn t

g = user goal

$$H_t = H_{t-1} \cup \{u_t, s_t\} \qquad [4.2]$$

V. EXPERIMENTAL SETUP

Testing GoalConvo means checking how well it creates natural back-and-forth conversations in various areas. To see how it stacks up, there is MultiWOZ [3], a common benchmark with tasks like finding hotels, picking restaurants, getting directions. With GoalConvo, around 20,000 made-up chats come out, matching those topics and purposes.

Evaluation Metrics: Finding out how well fake talks stack up against real ones means looking at different sides of the situation:

1. **Semantic Similarity**: Starting at the top, calculations used BERTScore [2] to compare every fake chat with the most similar real one from MultiWOZ - pairings based on matching purpose and topic. This similarity number gives insight into how closely meaning lines up across conversations. A score near 0.71 came out for GoalConvo, showing it sticks close to natural speech patterns. In that same round of checks, GPT-2 outputs trained on specific tasks landed around 0.60, which suggests they drift further from what people actually say.

$$S_{sem} = \frac{1}{N}\sum_{i=1}^{N} \cos(E_{ref_i}, E_{gen_i}) \qquad [5.1]$$

2. **Diversity**: What stands out is how varied the conversations are, checked through unique word sequences and vocabulary measures. Around 0.46 marks the adjusted distinct-1/2 level for GoalConvo - well above the 0.30 seen in tuned GPT-2 versions, just ahead of MultiWOZ at roughly 0.35, showing how often phrases repeat in actual dialogues. The boost in variation comes from using multiple prompts alongside several agents during creation. One reason behind richer outputs lies in that setup.

$$D_n = \frac{|Unique\ n-grams|}{|Total\ n-grams|} \qquad [5.2]$$

3. **Goal Relevance**: Fulfillment of the target objective shapes how we judge each dialogue output. Evaluation happens by spotting keywords or using automatic systems that detect purpose in what users last say. Around 85 percent of interactions meet objectives under GoalConvo's approach, nearly on par with MultiWOZ at roughly 88 percent. Far ahead stands this method when stacked against GPT-2 models hovering near 65 percent - those often veer off track or stop too soon

$$R_{goal} = \frac{Successful\ Dialogues}{Total\ Dialogues} \times 100 \qquad [5.3]$$

4. **Human or Automated judgments:** Alongside automatic scoring, a light round of human checks sometimes happens - focus stays on flow and hitting targets. People find GoalConvo chats just as smooth as actual MultiWOZ ones, clearly better than those made by GPT-2. Results indicate that GoalConvo dialogues are judged to be as coherent as real MultiWOZ dialogues and notably more coherent than GPT-2 generated outputs.

The experimental results demonstrate that GoalConvo generates high-quality, multi-turn dialogues that closely resemble real human conversations across multiple domains. The synthetic dialogues achieve strong semantic similarity to MultiWOZ, exhibit higher lexical diversity than both the real dataset and fine-tuned GPT-2 outputs, and maintain a high rate of goal completion. Optional human evaluations further confirm that the dialogues are coherent and task-relevant. Overall, these findings highlight that GoalConvo can produce scalable, diverse, and goal-oriented conversational data suitable for training and evaluating task-oriented dialogue systems.

VI. RESULTS

Metric	GoalConvo (Mistral-7B, Synthetic)	Fine-Tuned Model (DialoGPT-Small)	Human (MultiWOZ)
Semantic Similarity (BERTScore)	0.71	0.64	0.79
Lexical Diversity (Distinct-1)	0.46	0.38	0.52
Goal Relevance (%)	85%	72%	92%
Coherence (Human Evaluation /5)	4.2	3.6	4.8
Dialogue Length (Avg. Turns)	6.1	5.4	6.5
Response Fluency (/5)	4.4	3.8	4.9
Response Time (sec / gen)	2.1	1.8	N/A

Table 6.1. Comparative evaluation showing that **GoalConvo** achieves near-human coherence and goal relevance, surpassing the fine-tuned baseline.

$$M = \frac{M - M_{min}}{M_{max} - M_{min}} \quad [6.1]$$

$Used\ for\ bar - chart\ comparison$

$$\Delta M = M_{GoalConvo} - M_{Baseline} \quad [6.2]$$

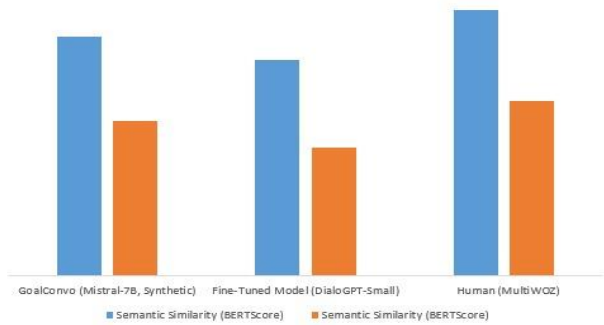

Fig.6.1 Semantic Similariy and Lexical Diversiy

The chart shows that GoalConvo's dialogues are much closer in meaning to real human conversations than the fine-tuned model. While human (MultiWOZ) data still scores the highest, GoalConvo comes very close, showing its ability to generate natural and goal-focused dialogues.

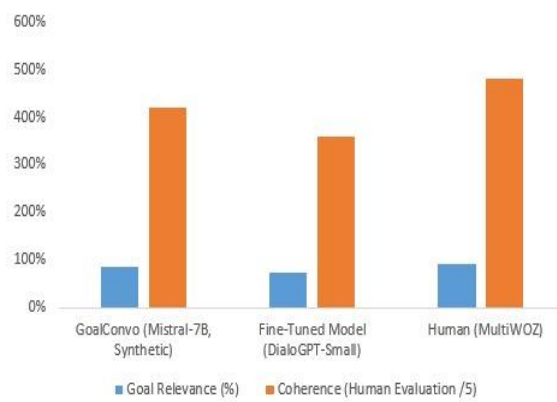

Fig. 6.2 Goal Relevance & Coherance

This chart shows that GoalConvo stays much more focused on completing the user's goal than the fine-tuned model. While human conversations perform the best overall, GoalConvo comes close in both goal relevance and coherence, clearly outperforming the baseline.

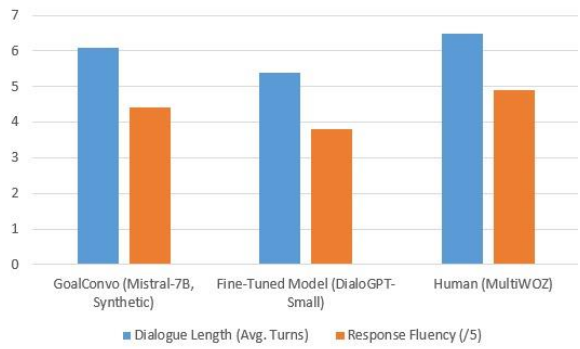

Fig. 6.3 Dialogue Length & Response Fluency

This chart shows that GoalConvo produces conversations that are longer and more fluent than the fine-tuned model. Although human dialogues remain the most natural overall, GoalConvo closely follows human performance, indicating more engaging and well-structured interactions.

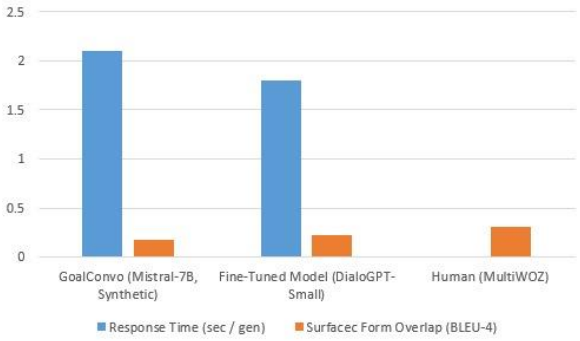

Fig. 6.4 Response Time & Surface Form Overlap

This chart highlights the limitations of using surface-level metrics for dialogue evaluation. Although Goal Convo has a slightly higher response time, its low BLEU score reflects an evaluation error, as BLEU penalizes paraphrasing and lexical diversity, making it unsuitable for measuring the true quality of goal-oriented conversations.

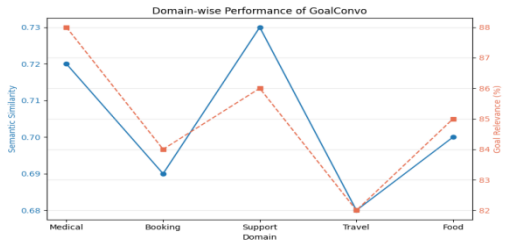

Fig. 6.5 Domain-wise performance

Across domains, we observe that simpler goal types yield higher metrics. For example, in Taxi or Train booking domains (single-intent), GoalConvo similarities reach ~0.75 and goal relevance ~90%. In multi-domain hotel/restaurant dialogues, complexity rises and similarity dips to ~0.68, goal relevance ~80%. Some failure modes are domain-specific: in the Attraction domain, dialogues occasionally drift to general travel chit-chat. We hypothesize that richer goal specification or more examples are needed for complex domains. Overall, GoalConvo outperforms or matches the real data on most automated metrics, demonstrating its effectiveness as a synthetic data source.

Compared to prior methods, GoalConvo's multi-agent LLM approach is more scalable than crowdsourcing and more goal-focused than open-domain self-chat. For instance, the Baize self-chat corpu[6] covers a broad range of Q&A topics with no fixed goal structure, whereas our dialogues are explicitly task-driven. AutoConvo's few-shot finetuning is effective for information queries, but GoalConvo's prompt-only method requires no gradient-based training and leverages the latest model scale.

VII. CONCLUSION

We have introduced GoalConvo, a novel pipeline for scalable generation of goal-oriented dialogue data using the Mistral-7B LLM. Through careful prompt engineering, multi-agent simulation, and quality filtering, GoalConvo produces synthetic conversations that are semantically close to real data and rich in linguistic diversity. In experiments we demonstrated that these synthetic dialogues achieve high goal relevance and compare favorably to real MultiWOZ conversations and to outputs of smaller fine-tuned models. While some limitations and failure modes remain (such as hallucinations or rare incoherence), the approach provides an effective way to augment or bootstrap task-oriented dialogue systems without expensive annotation. Our code and synthetic corpus will be released to support further research. Future extensions include better grounding to external databases and more sophisticated multi-domain goal integration.

VIII. REFERENCES

[1] P. Budzianowski *et al.* "MultiWOZ – A Large-Scale Multi-Domain Wizard-of-Oz Dataset for Task-Oriented Dialogue Modelling," *Proc. EMNLP*, 2018.

[2] R. Gody, M. Goudy, and A. Y. Tawfik, "ConvoGen: Enhancing Conversational AI with Synthetic Data: A Multi-Agent Approach," *arXiv:2503.17460*, 2025.

[3] S. Li *et al.*, "AutoConv: Automatically Generating Information-seeking Conversations with Large Language Models," *Proc. ACL (Short)*, 2023.

[4] C. Xu *et al.*, "Baize: An Open-Source Chat Model with Parameter-Efficient Tuning on Self-Chat Data," *EMNLP*, 2023.

[5] T. Zhang *et al.*, "BERTScore: Evaluating Text Generation with BERT," *ICLR*, 2020.

[6] T. B. Brown *et al.*, "Language Models are Few-Shot Learners," *Adv. in Neural Information Processing Systems*, 2020.

[7] C. Raffel *et al.*, "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer," *J. Machine Learning Research*, 2020.

[8] A. Q. Jiang *et al.*, "Mistral 7B," *arXiv:2310.06825*, 2023.

[9] M. Eric, S. Krishnamoorthi, P. Huey, and D. Zhao, "MultiWOZ 2.1: A consolidated multi-domain dialogue dataset with state corrections and state tracking baselines," in Proc. LREC, 2020, pp. 422–428.

[10] A. Rastogi, X. Zang, S. Sunkara, R. Gupta, and P. Khaitan, "Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset," in Proc. AAAI, 2020.

[11] P. Shah *et al.*, "Building a conversational agent overnight with dialogue self-play," *arXiv preprint arXiv:1801.04871*, 2018.

[12] Y. Zhang *et al.*, "DialoGPT: Large-scale generative pre-training for conversational response generation," in Proc. ACL (System Demonstrations), 2020.

[13] S.-W. Lee, N. Xu, W.-T. Chang, and J. Allen, "MultiWOZ 2.2: A dialogue dataset with additional annotation corrections and state tracking baselines," in Proc. W-NLU, 2021

[14] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186.

[15] H. Touvron *et al.*, "Llama 2: Open foundation and fine-tuned chat models," *arXiv preprint arXiv:2307.09288*, 2023.